

Attention-Based Retrieval vs Dense Semantic Search in RAG Systems

Aditya Sharma, 2026

1. Introduction

Retrieval-Augmented Generation (RAG) combines external knowledge retrieval with neural text generation. Traditional semantic search uses pre-trained sentence transformer models such as all-MiniLM-L6-v2 to map passages into dense embedding vectors. These models are typically trained on millions of sentence pairs across general domain corpora.

2. The Self-Attention Mechanism

Self-attention computes similarity weights across all token pairs in a sequence using query (Q), key (K), and value (V) matrices. By computing $\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V$, the model captures fine-grained contextual interactions between words.

A multi-head attention layer runs this computation in parallel across multiple representation subspaces, allowing the model to jointly attend to information at different positions.

3. Custom Lightweight Attention Encoders

Can a compact, custom-trained multi-head attention encoder trained with contrastive triplet loss compete with large pre-trained transformer backbones?

Contrastive learning pulls positive document chunks closer to the query while pushing hard negative distractor chunks farther apart in Euclidean space.

4. Empirical Evaluation Metrics

We evaluate retrieval performance using Recall@K and Mean Reciprocal Rank (MRR).

Recall measures the proportion of relevant chunks retrieved in top-K candidates.

MRR measures the reciprocal of the rank position of the first relevant chunk.

Latency benchmarks compare execution time across CPU and GPU inference environments.